

Worksheet: work out True or False before you run it

NAME _____

DATE Mon Sep 28 or Fri Sep 25 _____

CHUNK 4 OF 6

and: both must be True

and joins two conditions into one condition. The whole condition is True only when both sides are True. This is the chunk 4 program.

CHUNK 4, AS ON THE SESSION 2 PAGE

```
height = 52
has_ticket = True
if height > 48 and has_ticket:
    print("Board the coaster.")
else:
    print("Step aside.")
```

Each row of the table is one rider, as if lines 1 and 2 held that rider's values. In the two columns whose headings end in a question mark, write True or False. In the last column, write the line the program prints for that rider. Three rows are done for you, shaded, so you can check your working against them as you go. The rows in between are yours.

rider	height	has_ticket	height > 48?	height > 48 and has_ticket?	what it prints
1	52	True	True	True	Board the coaster.
2	52	False			
3	48	False			
4	48	True	False	False	Step aside.
5	40	True			
6	60	False			
7	49	False	True	False	Step aside.
8	49	True			

Look down your last column. Which riders board the coaster? Write their numbers, then one sentence: what do their rows have that no other row has?

Run it

Start from the program as written each time, and change one line per run. For rider 2, change line 2 to `has_ticket = False` . For rider 8, change line 1 to `height = 49` . Copy what it printed next to your prediction, and mark it right or wrong. Do not erase a wrong prediction.

Run	What I predicted	What it printed	Right?
rider 2			
rider 8			

CHUNK 5 OF 6

or: one is enough

`or` joins two conditions too. The whole condition is True when either side is True. This is the chunk 5 program.

CHUNK 5, AS ON THE SESSION 2 PAGE

```
is_staff = False
has_pass = True
if is_staff or has_pass:
    print("Gate opens.")
else:
    print("Gate stays shut.")
```

Each row is one visitor at the gate, as if lines 1 and 2 held that visitor's values. In the column whose heading ends in a question mark, write True or False. In the last column, write the line the program prints for that visitor. Visitor 1 is the program as written, done for you, shaded. Visitors 2, 3, and 4 are yours.

visitor	is_staff	has_pass	is_staff or has_pass?	what it prints
1	False	True	True	Gate opens.
2	True	False		
3	True	True		
4	False	False		

Which visitor does the gate keep out? Write the number, then one sentence: why does the gate stay shut for that visitor?

Run it

Start from the program as written. For visitor 4, change line 2 to `has_pass = False` and run it. Do not erase a wrong prediction.

Run	What I predicted	What it printed	Right?
visitor 4			

If you have time: and in place of or

On paper, change `or` to `and` on line 3. Which visitors would now get a different line printed? Write their numbers, then one sentence saying why.

A while loop and its stopping condition

A **while** loop checks its condition before every pass. When the condition is True, the block runs. The first time it is False, the loop stops and the line after the block runs.

!= means is not equal to. **""** is empty text: two quote marks with nothing between them.

CHUNK 6, AS ON THE SESSION 2 PAGE

```
answer = ""
while answer != "yes" and answer != "no":
    answer = input("Ready? Type yes or no: ")
print("Recorded:", answer)
```

Someone types maybe, then soon, then yes. Each row of the table is one check of the condition, in order. The answer column shows what **answer** holds at that check. In the three columns whose headings end in a question mark, write True or False. The last of them is the whole condition. Check 1 is done for you, shaded. Checks 2, 3, and 4 are yours.

check	answer	answer != "yes"?	answer != "no"?	answer != "yes" and answer != "no"?
1	""	True	True	True
2	"maybe"			
3	"soon"			
4	"yes"			

At every check where the whole condition is True, the block runs and the question appears. How many times does the question appear? Which check stops the loop, and which side of the **and** is False there?

Run it

Before you run it, write the lines you expect to see in the terminal, including what is typed after each question. Then run it, type maybe, soon, and yes, and copy each line. Do not erase a wrong prediction.

Line	What I predicted	What it printed	Right?
line 1			
line 2			
line 3			
line 4			

The same loop with or

Here is line 2 of the chunk 6 program with `or` in place of `and` . Every other line stays the same. Trace it on paper only. Someone types maybe, then yes, then no. Check 1 is done for you, shaded.

CHUNK 6, LINE 2 WITH OR

```
while answer != "yes" or answer != "no":
```

check	answer	answer != "yes"?	answer != "no"?	answer != "yes" or answer != "no"?
1	" "	True	True	True
2	"maybe"			
3	"yes"			
4	"no"			

Does the loop stop at check 3 or at check 4? Is there any answer that would make the whole condition False? One sentence for each question.

BUILD • 25 MINUTES

Build: the ride operator

The build asks for a height and a ticket answer. Write three of its lines here before you type them. Use `height` for the height, already changed to a whole number with `int()`, and `ticket` for the ticket answer.

Step 2 keeps asking until the rider types yes or no. Write the `while` line. It is the chunk 6 loop with `ticket` in place of `answer`.

In chunk 6, the line `answer = ""` comes before the `while` line. Write the line that comes before your `while` line. Then one sentence: what happens if it is missing?

Step 3 opens the ride only when the height clears 48 and the ticket answer is yes. Clears 48 means more than 48; exactly 48 goes to staff in step 4. Write the `if` line.
